

IE 376: Production Information Systems

Apparent Tardiness Cost (ATC) Dispatching Rule

For given scaling constant K , we take the ranking index for job j at time t as:

$$I_j(t) = \frac{w_j}{p_j} \exp \left(-\frac{\max\{d_j - p_j - t, 0\}}{K \cdot \bar{p}(t)} \right)$$

Where $\bar{p}(t)$ is the average of the processing times of the remaining jobs.

When the machine becomes free:

- Compute the index for all remaining jobs
- Select the job with the highest $I_j(t)$ value

Example . Single machine, total weighted tardiness problem with 4 jobs:

Jobs	1	2	3	4
p_j	13	9	13	10
d_j	12	37	21	22
w_j	2	4	2	5

Apply the ATC rule with $K = 5$.

Answer. Firstly, $\bar{p}(0) = \bar{p} = \frac{13+9+13+10}{4} = 11.25$

Moreover, at $t = 0$:

$$I_1(0) = \frac{2}{13} \exp \left(-\frac{\max\{12 - 13 - 0, 0\}}{5 \cdot 11.25} \right) = \frac{2}{13} = 0.1538$$

$$I_2(0) = \frac{4}{9} \exp \left(-\frac{\max\{37 - 9 - 0, 0\}}{5 \cdot 11.25} \right) = 0.2702$$

⋮

We continue the calculations and choose to schedule **Job 4** first (for full calculations refer to the table below).

Now, after Job 4 is processed, we are at $t = 10$. The remaining jobs are 1, 2, and 3.

$$\bar{p}(10) = \bar{p} = \frac{13+9+13}{3} = 11.67$$

$$I_1(10) = \frac{2}{13} \exp\left(-\frac{\max\{12 - 13 - 10, 0\}}{5 \cdot 11.67}\right) = \frac{2}{13} = 0.1538$$

$\vdots$

We proceed in the following manner and choose **Job 2** second. The full solution to the problem is as follows:

Time	<i>Ranking Index</i>	1	2	3	4	Job Scheduled
0	$I_j(0)$	0.1538	0.2702	0.1335	0.4039	Pick 4
10	$I_j(10)$	0.1538	0.3265	0.1538		Pick 2
19	$I_j(19)$	0.1538		0.1538		Pick 1
32						Pick 3

Under the schedule **4-2-1-3**, we have:

$$C_4 = 10, C_2 = 19, C_1 = 32, C_3 = 45$$

$$T_4 = 0, T_2 = 0, T_1 = 20, T_3 = 24$$

Hence, total weighted tardiness is $2 \cdot 20 + 2 \cdot 24 = 88$